

A VOLUME-LIMITED SEARCH FOR TECHNOSIGNATURES WITHIN 20 PC USING THE PUBLIC ALMA ARCHIVE

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ABSTRACT

We present a volume-limited search for radio technosignatures among the 120 stars within 20 pc with public ALMA archival coverage, selected solely on data availability – an unbiased sample covering only $\sim 5\%$ of the ~ 2400 catalogued stars within 20 pc. We reprocess archival visibilities to search simultaneously for narrowband Doppler-drifting technosignatures and anomalous continuum emission at $\sim 90\text{--}950$ GHz, frequencies largely unexplored by prior SETI work, supplemented by a closure-phase vetting test and a population-level continuum anomaly check, both unique to interferometric data. Of 120 planned target/bands, 21 are complete. No credible technosignature is found – one candidate is CO(1–0) emission – and we report EIRP limits of $1.6 \times 10^{13}\text{--}2.8 \times 10^{15}$ W (median 3.1×10^{14} W), five continuum detections, and sixteen non-detections to a median 5σ limit of ~ 0.55 mJy. The survey is ongoing.

Subject headings: extraterrestrial intelligence — radio lines: stars — submillimetre: stars — surveys — astrobiology

1. INTRODUCTION

The search for extraterrestrial intelligence (SETI) rests on the premise that a technological civilisation, whether deliberately signalling or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Since the foundational proposal of Cocconi & Morrison (1959) that microwave frequencies near the 21 cm hydrogen line would be an obvious “meeting place” for interstellar communication, and Frank Drake’s first observational test of the idea with Project Ozma (Drake 1960), the radio band has remained the dominant search space for technosignatures, for reasons that are both historical and physically well motivated: radio photons are cheap to produce in enormous numbers, propagate through the interstellar medium and planetary atmospheres with little attenuation, and (for a sufficiently narrowband signal) can in principle be distinguished from any known natural astrophysical process by their spectral compactness and Doppler dynamics (Sheikh 2020).

Six decades on, radio technosignature searches have grown from single-target, single-frequency pointings into surveys of many thousands of stars across broad swathes of the radio spectrum. Yet, as Wright et al. (2018) demonstrated quantitatively with their “Cosmic Haystack” formalism, even the combined efforts of every major SETI programme to date have sampled only an infinitesimal fraction of the plausible multi-dimensional parameter space of transmitter frequency, bandwidth, power, location, and duty cycle. Persistent, non-trivial gaps remain: almost the entire millimetre and submillimetre regime above ~ 50 GHz is essentially unsearched; most surveys have targeted specific classes of star (nearby exoplanet hosts, young stellar objects with protoplanetary discs, stars in the “Earth Transit Zone”) rather than an unbiased, complete sample of the local stellar population; and only a small number of studies have systematically reprocessed *existing* interferometric archives for technosignatures, as opposed to requesting dedicated telescope time.

This paper describes a new survey designed to address several of these gaps simultaneously. We construct a volume-limited sample of every star within 20 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archival data for both narrowband technosignature candidates and anomalous continuum emission, applying corrections for stellar proper motion and for the range of Doppler drift rates plausible for a transmitter on a planet in orbit around its host star. We further supplement this classical search with two analyses that exploit properties specific to this survey and unavailable to prior technosignature work: interferometric closure-phase vetting of any candidate signal, and a population-level anomaly check enabled by the sample’s unbiased, volume-limited design. Section 2 reviews the relevant literature: the history of radio SETI (§2.1), the current generation of large-scale technosignature surveys (§2.2), the very limited body of work at millimetre/submillimetre frequencies (§2.3), the growing use of machine learning and anomaly detection in this field (§2.4), the statistical frameworks used to interpret survey non-detections (§2.5), the case for a volume-limited, representative sample (§2.6), and the prospects offered by next-generation facilities (§2.7). Section 3 summarises our sample selection, Section 4 our data-reduction and search methodology (including the two new analyses), Section 5 presents our results to date, and Section 6 discusses their interpretation and the survey’s prospects. Full per-target tables are deferred to Appendix A so as not to interrupt the main narrative with tabular material.

2. BACKGROUND

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2.1. Six decades of radio SETI: a brief history

Project Ozma (Drake 1960) observed two nearby Sun-like stars, τ Ceti and ϵ Eridani, with the 26 m Tatel telescope at Green Bank at 1420 MHz for a total of some 150 hours, and found no signal, but established both the methodology and the symbolic starting point for the modern field. Through the 1960s–1980s a steady stream of targeted and sky-survey searches followed, including Ozma II (Drake 1986), the Ohio State “Big Ear” survey (source of the famous “Wow!” signal), and the META and BETA all-sky surveys conducted from Harvard/Smithsonian facilities. Project Phoenix (Backus & Project Phoenix Team 2002), run by the SETI Institute from 1995–2004, was the most sensitive targeted survey of its era, observing on the order of 800 nearby Sun-like stars using the Arecibo, Parkes and Green Bank telescopes, and pioneered a rigorous methodology for distinguishing candidate signals from terrestrial radio-frequency interference (RFI) through simultaneous multi-site or multi-antenna verification. The Allen Telescope Array (ATA; Tarter 2001), jointly developed by the SETI Institute and UC Berkeley and first commissioned in 2007, was the first radio telescope purpose-built from the outset for commensal, wide-band SETI observing, and remains in active use, including in recent technosignature searches of interstellar objects (Sheikh et al. 2025b) and TRAPPIST-1 (Tusay et al. 2024). The distributed-computing project SETI@home (Von Korff et al. 2025) processed public-participant-donated CPU cycles to search Arecibo sky-survey data for narrowband and pulsed signals for two decades before its 2020 shutdown, and its final data-analysis summary was only published in 2025.

Two complementary observing strategies have been used throughout this history: *sky surveys*, covering a large fraction of the celestial sphere at modest sensitivity per pointing, and *targeted searches*, concentrating integration time on a preselected list of promising stars (e.g. Drake 1974). Both strategies persist in the modern era, but – as we discuss in §2.6 – the majority of targeted programmes have selected their samples on some presumed indicator of habitability or interest (proximity, Sun-like spectral type, known planets, or membership of the Earth Transit Zone), rather than aiming for a complete, unbiased census of the local stellar population.

2.2. Modern large-scale radio technosignature surveys

The current generation of technosignature science is dominated by the Breakthrough Listen initiative, launched in 2015 with a ten-year, \$100 million commitment to survey the one million nearest stars, the full Galactic plane and centre, the 100 nearest galaxies, and a sample of exoplanet and planet-candidate hosts, primarily using the Robert C. Byrd Green Bank Telescope (GBT) and the CSIRO Parkes “Murriyang” telescope, and now increasingly via commensal observing on MeerKAT (Isaacson et al. 2017). Its first major published dataset, Enriquez et al. (2017), searched 692 nearby stars over 1.1–1.9 GHz with the GBT and set limits on narrowband transmitters of $\text{EIRP} > 10^{13} \text{ W}$; this was followed by Price et al. (2020) (1327 stars, 1.10–3.45 GHz), Gajjar et al. (2019)-style searches of the restricted Earth Transit Zone at 3.95–8.00 GHz, and, most recently, a substantial expansion of automated, commensal technosignature searching with MeerKAT that has processed some 1.5 million coherent beams between 2023 and 2026 (Czech et al. 2026). In parallel, Margot et al. (2023) conducted a large citizen-science-style search of 11 680 stars with the GBT at 1.15–1.73 GHz, explicitly developing a statistical formalism for interpreting non-detections in terms of the fraction of stars that could host a detectable transmitter, *provided the observed sample is representative of the underlying stellar population of the search volume* – a condition that, as we discuss below, is often not satisfied by targeted, interest-selected samples.

Beyond Breakthrough Listen, the last five years have seen international radio facilities extend technosignature searches to new frequency regimes and observational strategies: the Five-hundred-metre Aperture Spherical Telescope (FAST; Zhang et al. 2020; Li et al. 2026) in China has conducted commensal searches with unmatched raw sensitivity at L-band, including recent narrowband searches of interstellar objects; LOFAR and the pathfinder NenuFAR have pushed technosignature searches down to 10–190 MHz, with LOFAR alone having surveyed over 1.6 million stars in commensal mode and dual-site LOFAR configurations enabling near-real-time RFI rejection (Johnson et al. 2023); the Sardinia Radio Telescope has conducted one of the first dedicated high-frequency ($> 18 \text{ GHz}$) surveys of nearby stars and the Galactic Centre (Valente et al. 2025); and commensal technosignature back-ends are now operating or in development at the Karl G. Jansky Very Large Array (the COSMIC system; Tremblay et al. 2024), MeerKAT (Czech et al. 2021), and the Murchison Widefield Array (Morrison et al. 2023). Occultation-based strategies – observing a star during a planet-planet or planet-star occultation, when geometric alignment transiently favours detection of leakage or deliberate signals – have also grown in popularity, exemplified by recent Breakthrough Listen searches of TESS-selected eclipsing systems (Tusay et al. 2025).

2.3. Millimetre and submillimetre technosignature searches

In stark contrast to the radio regime below $\sim 10 \text{ GHz}$, the millimetre and submillimetre bands remain almost entirely unexplored for technosignatures. Mason et al. (2024) conducted, to our knowledge, the first technosignature search ever performed with ALMA, searching archival Band 3 data (spectral windows centred at 90.642 and 93.151 GHz) for narrowband signals towards 28 Galactic stars selected from Gaia DR3. Their approach used a “stellar bycatch” method: rather than targeting stars directly, they searched for serendipitous stars falling within the undistorted field of view of four bright ALMA phase calibrators, since these calibrators are observed frequently and are not generally themselves of astrophysical interest. For the nearest star in their sample they placed a limit of $\text{EIRP}_{\text{min}} > 7 \times 10^{17} \text{ W}$, and discussed in detail the specific challenges of high-frequency technosignature work: substantially larger Doppler drift rates at fixed line-of-sight acceleration (since drift rate scales linearly with observing frequency), the resulting sensitivity loss (“smearing”) for a fixed spectral and temporal resolution unless a drift-rate search is performed, and

an increased incidence of spectral confusion from the dense molecular line forest present in protoplanetary discs and star-forming regions at these frequencies. Tusay et al. (2024) extended technosignature work on the TRAPPIST-1 system to the Allen Telescope Array, underlining the continued observational interest in this particular target but also its status as one of very few systems that has now been searched for technosignatures across a wide range of frequencies. Our own earlier work (White 2026) reproduced and extended the Mason et al. (2024) methodology directly on ALMA science-target (rather than calibrator-bycatch) fields for TRAPPIST-1 across Bands 3, 6 and 7, finding no detection and placing limits of $\text{EIRP}_{\min} \sim 5\text{--}10 \times 10^{13} \text{ W}$, and subsequently built a ranked target list and pilot survey of 100 nearby, ALMA-observed stars, from which the present, volume-limited survey has grown. Beyond these efforts, we are not aware of any other published, systematic technosignature search using ALMA or any comparably sensitive millimetre/submillimetre interferometer, representing a substantial and readily addressable gap in frequency coverage relative to the sub-10 GHz literature.

The near-complete absence of mm/submm technosignature work to date is attributable less to any fundamental unsuitability of these frequencies – Sheikh’s “nine axes of merit” framework (Sheikh 2020) notes that higher frequencies suffer less from interstellar scintillation and dispersion, and offer access to much larger instantaneous bandwidths – than to historical circumstance: sensitive mm/submm interferometers with the necessary spectral resolution and archival depth (chiefly ALMA) have only existed, and only accumulated a sufficiently large calibrated science archive, within roughly the last decade. This makes archival reprocessing, rather than dedicated proposal-driven observing, a particularly natural strategy in this regime: ALMA’s archive already contains many thousands of hours of high-spectral-resolution observations of individual, nearby stars, taken for entirely unrelated astrophysical purposes (disc structure, molecular chemistry, stellar activity, debris-disc detection, calibration monitoring), each of which can in principle be re-searched for technosignatures at negligible additional observing cost.

2.4. Machine learning and anomaly detection in technosignature searches

The scale of modern technosignature datasets – hundreds of hours of high-time- and frequency-resolution data per survey, often containing many thousands of candidate “hits” dominated by anthropogenic RFI – has made machine learning an increasingly central tool for both signal detection and RFI rejection. Ma et al. (2023) presented the first large-scale deep-learning technosignature search, applying a β -convolutional variational autoencoder to 480 hours of Breakthrough Listen GBT data on 820 nearby stars in a semi-unsupervised anomaly-detection mode, recovering candidate signals missed by the standard TURBOSETI narrowband drift-search pipeline (Enriquez & Price 2019) and reducing the number of signals requiring human vetting by roughly two orders of magnitude, while returning eight signals of interest for re-observation. Building on this, Choza et al. (2024) identified and corrected a long-standing sensitivity mischaracterisation in TURBOSETI’s signal-to-noise calibration, prompting a retrospective correction of quoted sensitivities in several earlier Breakthrough Listen publications (Lacki et al. 2021). More recently, unsupervised clustering methods (e.g. the “GLOBULAR” algorithm) have been developed specifically to separate genuine anomalies from RFI in narrowband technosignature searches without requiring labelled training examples (Jacobson-Bell et al. 2025), and further anomaly-detection searches have been applied to combined Parkes and GBT Breakthrough Listen archives (Poznanski et al. 2025). These efforts collectively demonstrate that machine-learning-based anomaly detection can recover classes of signal (broadband, non-linearly drifting, or otherwise morphologically unusual) that narrowband, linear-drift matched-filter searches such as TURBOSETI are, by design, insensitive to (Garrett et al. 2026), and that a purely classical, threshold-based search – of the kind we adopt for the present survey, following Mason et al. (2024) – necessarily leaves some fraction of the true technosignature parameter space unexamined. We return to this point in §6 as a natural direction for follow-up work once the present survey’s baseline dataset exists.

2.5. Statistical frameworks for interpreting non-detections

A non-detection is not, by itself, a null result: it must be translated into a quantitative statement about which regions of transmitter parameter space have been excluded, and with what confidence. Wright et al. (2018) formalised this with the “Cosmic Haystack” framework, modelling the technosignature search space as an eight-dimensional volume (spanning, e.g., transmitter frequency, bandwidth, location, distance, and duty cycle) and showing that even the combined efforts of every major SETI programme conducted up to 2018 sampled a small fraction of any astrophysically plausible haystack volume – a result frequently paraphrased as showing that humanity has searched “the equivalent of a large hot tub’s worth of the Earth’s oceans”. Sheikh (2020) subsequently proposed a complementary “nine axes of merit” rubric for comparing the relative scientific value of different technosignature search strategies (spanning axes such as detectability, transmitter cost and information content), providing a qualitative framework alongside Wright et al.’s quantitative one. Margot et al. (2023) developed an explicit formalism for converting a survey’s non-detection rate into an upper bound on the fraction of stars in the search volume hosting a detectable transmitter, but noted explicitly that this inference is only valid if the observed sample is statistically representative of the underlying stellar population of the search volume – a requirement that most targeted, interest-selected samples (including our own earlier 100-star pilot list) do not, by design, satisfy. Most recently, Sheikh et al. (2025a) inverted the usual question, asking at what distance Earth’s own technosignatures would themselves be detectable by an ALMA- or SKA-like observatory, providing a valuable calibration point against which the sensitivity of any given survey, including the present one, can be benchmarked.

2.6. The case for a volume-limited, representative sample

The great majority of the surveys reviewed above – including our own earlier 100-star ALMA pilot programme – select targets by some combination of proximity, Sun-like spectral type, known exoplanets, protoplanetary-disc or debris-disc detection, or membership of specially-constructed target lists such as the Earth Transit Zone. Such selections are entirely reasonable when the goal is to maximise the a priori probability of detecting a genuine technosignature per unit observing time, but they carry an unavoidable cost: they make it difficult to convert a survey’s results into a statistically clean statement about the prevalence of technosignatures in the general stellar population, precisely because the sample is not representative of that population (Margot et al. 2023). A disc-hosting or exoplanet-hosting bias is particularly worth flagging in the ALMA context: ALMA-detectable circumstellar and protoplanetary discs are, almost without exception, orders of magnitude brighter than the present-day Solar System’s own zodiacal dust cloud would appear at interstellar distances, so that samples selected *because* they have an ALMA-detected disc are biased towards young, dynamically active systems that are not necessarily more (and could plausibly be less) likely to host a mature, Earth-like technological civilisation than an old, quiescent, disc-poor star of the same spectral type.

The present survey is explicitly designed to avoid this selection effect. We construct our sample by identifying every star within 20 pc of the Sun that happens to fall within the field of view of at least one public ALMA observation – irrespective of whether that observation was proposed to study a disc, a stellar flare, an unrelated background source, or anything else – and irrespective of the target star’s own spectral type, age, multiplicity, or known planet-hosting status. The star need not be at the pointing centre of the observation, provided it lies within the usable primary-beam field of view (see §3). This is, in effect, a volume-limited “bycatch” census of the ALMA archive: rather than sampling stars selected for astrophysical interest in the calibrator fields as in Mason et al. (2024), we sample every nearby star that ALMA happens to have observed for any purpose whatsoever, within a fixed physical volume. We recognise, and quantify explicitly in §3 (finding that our 120-star sample represents only $\sim 5\%$ of the known stellar population within 20 pc), that this approach is still subject to a residual, second-order selection effect – ALMA archival coverage of the immediate solar neighbourhood is itself not spatially uniform, since some regions and spectral types are more likely to have been proposed for ALMA time than others – but this bias operates on the presence of *any* ALMA coverage of a star, not on the astrophysical properties that are the actual subject of a technosignature search, and can therefore be characterised and reported as a well-defined completeness function of the 20 pc volume, rather than an implicit and unquantified bias in the target selection itself.

2.7. Prospects for next-generation facilities

The coming decade will see a substantial expansion of technosignature search capability. The Square Kilometre Array (SKA; Siemion et al. 2015; Tremblay et al. 2026), now under construction in Australia (SKA-Low) and South Africa (SKA-Mid), will combine wide instantaneous bandwidth, high angular resolution, and an architecture well suited to commensal, multi-beam technosignature searching, with projected sensitivity sufficient to detect Earth-analogue-level radar and communication leakage from significantly greater distances than is possible today. The next-generation Very Large Array (ngVLA; Ng et al. 2022) is specifically being designed with technosignature science as one of its commensal use cases, with proposed antenna configurations optimised to allow of order 64 simultaneous high-spectral-resolution commensal beams. Existing commensal systems at MeerKAT, the VLA (COSMIC), and the Murchison Widefield Array are already providing a preview of the observing modes these larger facilities will employ at scale. None of these facilities, however, offer ALMA’s combination of angular resolution, spectral resolution and raw collecting area above ~ 50 GHz in the near term, so a systematic mm/submm technosignature census – of the kind begun here – is likely to remain a uniquely ALMA-enabled measurement for some years yet, and represents a natural complement to the lower-frequency, larger-sample surveys that will be enabled by SKA and ngVLA.

3. SAMPLE SELECTION

Our sample comprises every star within 20 pc of the Sun (based on Gaia DR3 parallaxes) with at least one public ALMA observation, of any band, science category, or proposal type, in which the star’s true (proper-motion-corrected) position at the epoch of observation falls within the field of view of at least one execution block – not necessarily at the observation’s pointing centre, since a substantial fraction of ALMA mosaics and wide single-pointing observations serendipitously cover more than one catalogued star. We deliberately do not weight target selection by the presence of a circumstellar disc, known exoplanets, spectral type, or any other astrophysical property beyond distance and archival data availability; disc presence and planet-hosting status are retained purely as descriptive metadata for the final sample, not as selection or ranking criteria. At the time of writing this identifies 120 unique target stars, spanning spectral types from early A (e.g. Sirius B, α Cma B) through to the latest, faintest M dwarfs (e.g. Proxima Centauri, TRAPPIST-1, Barnard’s Star, Teegarden’s Star), ranked by increasing distance from the Sun so that the nearest, most sensitively-constrained systems are processed first. The underlying pipeline is built to extend this volume-limited approach to 30, 40 or 50 pc as a natural next step, at the cost of a rapidly increasing target list and a correspondingly reduced fraction with existing ALMA coverage, since the archive’s areal completeness at fixed exposure declines with distance.

It is important to be explicit about what “volume-limited” does and does not mean here: our 120 targets are every star within 20 pc with public ALMA coverage, *not* a complete census of every known star within 20 pc. Cross-matching against our own reference Gaia DR3 catalogue of the solar neighbourhood, 2357 individual stars are catalogued within 20 pc, of which our 120 targets represent only $\sim 5\%$ (Figure 2). The sample is volume-limited, and unbiased with respect to spectral type, disc-hosting status, and planet-host status, *within the population that ALMA happens to have observed for any purpose*; it is not, and does not claim to be, an unbiased sample of the local stellar population as a

whole, since the decision to propose ALMA time towards a given nearby star is itself not uniformly distributed across spectral type or other stellar properties. We regard this as a genuine, quantifiable limitation of an archival-mining approach, worth stating plainly rather than allowing the term “volume-limited” to be read as “complete”.

4. DATA AND METHODOLOGY

For each target we mine the public ALMA science archive for all calibrated (or, where necessary, locally recalibrated from raw data) execution blocks covering the star’s position, apply an efficiency-bounded download and processing budget per target to remain within the resource constraints of the shared compute facility used for this work, and search the resulting visibility data in two complementary ways: (i) a narrowband, Doppler-drift-corrected search for technosignature candidates, following the general methodology of Mason et al. (2024) and Enriquez et al. (2017), converting any non-detection into an equivalent isotropic radiated power (EIRP) limit via $\text{EIRP}_{\min} = 4\pi d^2 S_{\min} \Delta\nu$; and (ii) a line-excluded continuum search for anomalous broad-band emission at the star’s position, with an accompanying epoch-to-epoch variability check wherever multiple execution blocks exist. Stellar positions are propagated to the epoch of each observation using Gaia DR3 proper motions and parallaxes. The Doppler-drift search range applied to each target is set to the larger of (a) a generic frequency-scaled default following Mason et al. (2024)’s adoption of a literature fiducial with a threefold safety margin, and (b) where the target hosts one or more confirmed exoplanets, an explicit calculation of the maximum line-of-sight orbital acceleration (and hence maximum plausible transmitter drift rate) of the innermost known planet, so that a transmitter co-located with a known, short-period planet cannot be missed simply because its orbital motion drifts it out of an insufficiently wide search window. This same drift-rate search comfortably subsumes any drift induced by the Earth’s own orbital motion around the Sun (a peak line-of-sight acceleration of $5.9 \times 10^{-3} \text{ m s}^{-2}$, corresponding to $\sim 0.02 \text{ Hz s}^{-1}$ per GHz), more than two orders of magnitude below the 12 Hz s^{-1} per GHz generic default, so no separate barycentric correction is required. The trial drift-rate grid itself depends only on the elapsed time and channel width of the observation, not on sky position, and is applied identically to the star and to every control/comparison position used for false-alarm calibration (§4.2): the frequency-shift correction therefore has no dependence on the target’s position relative to the observation’s phase centre.

It is important to be explicit about the fraction of each observation’s instantaneous bandwidth that the narrowband search actually covers. ALMA’s correlator is typically configured with several spectral windows observed simultaneously (a mix of coarse, wide-bandwidth continuum windows and one or more finer-resolution windows, or, for spectral-scan observations, many narrow windows tuned across a wide frequency range). Early versions of this survey, following Mason et al. (2024), searched only the single finest-resolution spectral window available for each target, since narrowband sensitivity scales with channel width. We have since extended the pipeline to search *every* distinct spectral window configured for a given observation (up to a practical cap of 8 windows per target, to bound compute cost on the shared cluster used for this work; §3), each yielding its own independent $\text{EIRP}_{\min}$ limit and closure-phase vetting outcome (§4.2), reported as a separate row of Table 1 (Appendix A). Of the 21 target/bands processed to date, only GJ 581 has so far had every one of its configured windows searched; the remaining 20 are being reprocessed under the extended pipeline as the survey continues, and Table 1 still shows only their originally-searched window, explicitly flagged, until reprocessing completes. Across all target/bands processed to date under either pipeline version, the number of spectral windows configured simultaneously by the correlator ranges from 4 to 24 (median 4). For Figure 1, to avoid crowding the plot with multiple points per target once several windows have been searched, we plot only the single deepest (lowest $\text{EIRP}_{\min}$) result per target/band. The line-excluded continuum search (ii), by contrast, already used the full available bandwidth across all configured spectral windows for each target in every version of the pipeline, since it does not require narrowband spectral resolution.

Any narrowband hit that crosses the detection threshold is automatically cross-checked against the rest frequencies of a set of common mm/submm molecular transitions (e.g. the CO, HCN, HCO^+ , CS and CN lines already used to exclude line channels from the continuum search) at the star’s own frame, and is flagged as probable line contamination, rather than a technosignature candidate, if its recovered frequency lies within one channel width of a known transition *and* its best-fit drift rate is consistent with zero – a genuine spectral line is stationary in frequency over the observation, whereas the search is explicitly looking for *drifting* signals, so this combination of frequency coincidence and negligible drift is what distinguishes line contamination from a real drifting candidate. This check is applied uniformly to every spectral window of every target, and is how the one hit produced by the survey to date (β Pictoris, §5) is identified as CO(1–0) emission.

We also confirm that both the narrowband EIRP limits and the continuum flux densities/upper limits reported here are corrected for ALMA’s primary-beam response. The star’s true (proper-motion-corrected) sky position is not, in general, exactly coincident with the phase centre of the ALMA field it is being searched against; the interferometric response to a source away from the phase centre is attenuated by the primary beam, and both pipelines now explicitly divide the measured flux/rms by the (Gaussian-approximated) primary-beam power response at the star’s actual offset before it is reported. For the great majority of targets processed to date this correction is negligible (offsets of $\lesssim 2''$ against primary beam FWHMs of $17\text{--}70''$, a $<0.3\%$ effect, including for the UV Ceti pair, whose own $\sim 2''$ offset gives a 0.2% correction), but for the highest-proper-motion target searched so far, Sirius B, where the star’s position has moved $\sim 7\text{--}8''$ relative to the ALMA pointing between the archival observation’s epoch and the star’s current Gaia-propagated position, the correction reaches $3.6\text{--}16\%$ depending on band (Table 1); the values quoted for Sirius B include it.

4.1. Reference transmitter benchmarks

To give the EIRP limits reported in §5 physical context, we compare them throughout to two benchmark values for real, existing human transmitters, both taken from Sheikh et al. (2025a): an Arecibo-like planetary radar, whose characteristic S-band EIRP was 2×10^{13} W (the strongest transmitter ever built by humans, used historically both for Solar System radar astronomy and for the 1974 Arecibo Message); and typical, entirely unintentional Earth radio leakage from mobile/cellular infrastructure, with an EIRP of order 4×10^9 W. These two values bracket a physically motivated range: the former represents what a civilisation with our own most powerful *deliberate, directed* transmitter could achieve if pointed at us; the latter represents what an entirely unintentional, Earth-2024-equivalent level of technological activity would radiate in our direction as background leakage, with no deliberate signalling intent at all.

4.2. Closure-phase point-source vetting

Any narrowband signal that crosses our detection threshold at the star’s position is subjected to a further, automated test that is, to our knowledge, unavailable to any single-dish or phased-array-beamformed technosignature survey (Breakthrough Listen, FAST, the ATA, LOFAR, and similar all lack it, since it requires multiple independent baselines): the interferometric closure phase, the phase sum $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a closed triangle of baselines i - j - k . This quantity is a standard radio-interferometric identity that is theoretically *exactly zero* for an unresolved point source at *any* sky position – a point source’s visibility phase is a pure linear function of (u, v) , which cancels identically around any closed baseline loop – and is, by construction, immune to per-station calibration errors. A candidate signal whose closure phase is statistically inconsistent with zero at the drift-tracked frequency of the hit is therefore evidence against it behaving as a genuine celestial point source, favouring instead a near-field, terrestrial, or instrumental origin. This test is independent of, and complementary to, the existing spatial-coincidence test (whether the on-source position shows a larger peak than the control positions): closure phase does not depend on where the source is, so it answers a different question – “does this behave like a coherent point source at all” – than the spatial test’s “is it located at the star”. We validated the closure-phase calculation against synthetic visibility data before applying it to any ALMA measurement, confirming zero closure phase for a noiseless simulated point source, statistical consistency with zero for a noisy simulated point source, and a large, clearly inconsistent scatter for a simulated non-point-source (RFI-like) signal. Because it is only triggered on an actual threshold-crossing candidate, this test adds negligible computational cost to the survey as a whole.

4.3. Population-level, spectral-type-normalised continuum anomaly check

Because our sample is volume-limited and not selected on disc-hosting or planet-host status, it supports a comparison that is not meaningful for the biased target lists used in essentially every prior technosignature survey: whether a star’s measured continuum brightness is anomalous relative to the physically-expected flux of its own stellar photosphere, and relative to other stars of similar effective temperature already processed within the same survey. For each target we compute the expected photospheric flux density from the Planck function and the star’s own solid angle, $S_\nu = \Omega \times 2h\nu^3/c^2 \times [\exp(h\nu/k_B T_{\text{eff}}) - 1]^{-1}$ with $\Omega = \pi R_\star^2/d^2$, using Gaia DR3 photometric temperatures and, where available, Gaia GSP-Phot radii (supplemented by a coarse Pecaut & Mamajek main-sequence T_{eff} -radius relation where a direct Gaia radius estimate is unavailable, as is the case for close to half of our sample – Gaia’s astrophysical-parameter pipeline has known gaps for the very bright/nearby and very red/faint stars that a 20 pc sample is disproportionately composed of). Any large excess of the measured continuum flux (or, for a non-detection, its 5σ upper limit) above this photospheric baseline is flagged, and is additionally compared against the distribution of excess ratios already measured for other stars in the same coarse T_{eff} bin, once at least five such comparison stars exist, using a robust (median/MAD-based) outlier statistic. This check requires no additional raw-data retention beyond what the continuum measurement itself already produces, and updates incrementally as the survey proceeds.

5. RESULTS

As of this version of the paper, 21 of the planned 120 target/bands have been searched to completion (a further 2, Barnard’s Star and Wolf 359, completed their continuum measurement but not the narrowband search, owing to a data-quality failure in those specific execution blocks; both are flagged for an automatic, non-gating retry). A further 4 target/bands that completed processing during this update are deliberately excluded from the results below, having been found by an automated pointing sanity check to be associated with the wrong ALMA dataset entirely (§4, Appendix A); these are being re-investigated rather than reported. The survey is strictly volume-ordered (nearest star first; §3), so the results reported here are weighted towards the very nearest stars in the sample, spanning 1.3–19.6 pc at the time of writing. Full per-target values, including the exact frequency range searched in each band, are given in Table 1 in Appendix A rather than in the main text; this section summarises the results and highlights points of interest.

5.1. Technosignature search

Nineteen of the 21 target/bands yielded at least one narrowband search result (Barnard’s Star and Wolf 359 failed narrowband extraction outright and carry no EIRP limit at this time); together these span 22 individual spectral-window measurements, since GJ 581 now has a result for all four of its configured windows (§4). The resulting EIRP_{min} limits span 1.6×10^{13} – 2.8×10^{15} W, with a median of 3.1×10^{14} W (Figure 1, left panel); as expected for a volume-ordered survey, the deepest limits are for the very nearest systems (e.g. 1.7×10^{13} W for the UV Ceti binary at ~ 2.7 pc).

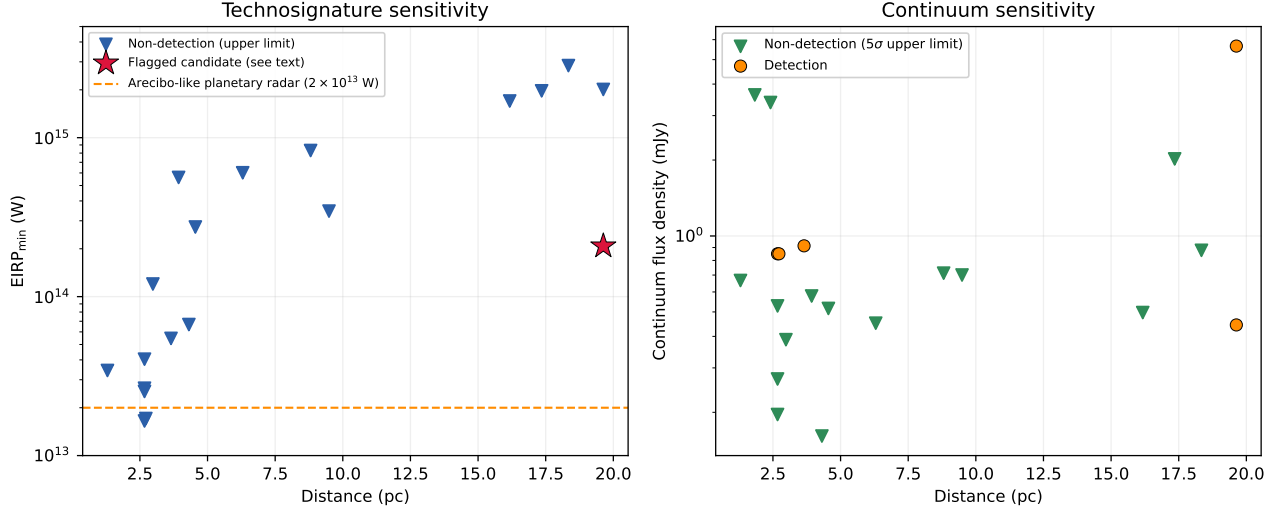


FIG. 1.— Technosignature EIRP upper limits (left) and continuum flux densities (right; downward triangles are 5σ upper limits, i.e. $5\times$ the map rms noise – see §5 and Table 1 for the rms values themselves – filled circles are detections) as a function of distance, for the 21 target/bands searched to date. The single automatically-flagged candidate (β Pictoris, Band 3) is marked with a star and is identified in the text as CO(1–0) line contamination rather than a technosignature. The left panel also marks, as a reference line, the EIRP of an Arecibo-like planetary radar (2×10^{13} W, the most powerful deliberate transmitter yet built by humans; Sheikh et al. 2025a, §4.1). A second benchmark from the same reference, typical unintentional Earth radio leakage ($\sim 4 \times 10^9$ W), is discussed in the text (§5) but is not plotted here, since it lies more than three decades below every point in this panel and including it would force the y -axis to span an unusably large range for comparing the points themselves.

Compared against the reference transmitter benchmarks of §4.1, our very deepest limits (the handful of targets within ~ 3 pc) are only just below the 2×10^{13} W Arecibo-like planetary-radar benchmark, meaning we could have detected an equally powerful, deliberately Earth-directed transmitter at those distances had one been present; every target beyond ~ 4 pc, and every target with respect to the much fainter $\sim 4 \times 10^9$ W level of Earth’s own unintentional radio leakage, remains well out of reach of the present survey’s sensitivity (Figure 1). This is a useful, sobering calibration: even this deep, targeted, archival mm-wave search is not yet sensitive to an Earth-equivalent civilisation’s background leakage at any of the distances probed so far, and is only marginally sensitive to an Earth-equivalent civilisation’s most powerful deliberate transmitter, and only for its very nearest neighbours. One narrowband hit was automatically flagged by the pipeline as a “credible technosignature candidate” (on-source peak signal-to-noise ratio above threshold and localised to the star’s position, rather than appearing equally in the control positions): a signal towards β Pictoris in ALMA Band 3 at 115.26 GHz. This frequency coincides, to within the spectral resolution of the observation, with the rest frequency of the CO($J = 1 \rightarrow 0$) rotational transition (115.271 GHz), and β Pictoris is a well-known debris-disc system with previously reported CO emission; we therefore attribute this hit to line contamination rather than a genuine technosignature. No other candidate has survived both the spatial-coincidence and (where triggered) closure-phase vetting tests described in §4.2; in practice, the closure-phase test has not yet been exercised against a real ALMA measurement, since no other candidate has crossed the initial detection threshold to trigger it.

5.2. Continuum search

Of the 21 target/bands with a continuum measurement, five show a $\geq 5\sigma$ continuum detection and sixteen are non-detections (Figure 1, right panel). Throughout this paper, a continuum “upper limit” is explicitly $5\times$ the map rms noise at the star’s position (i.e. the same 5σ threshold used to define a detection in the first place), primary-beam corrected exactly as the EIRP limits are (§4); the underlying rms values themselves are also given per target in Table 1 so that a reader wanting a different confidence level can rescale directly, rather than being limited to a single quoted multiple of σ . On this basis the non-detections reach 5σ upper limits of 0.16–3.6 mJy (median 0.55 mJy). Two of the five detections – catalogued separately in our target list as G 272-61A and G 272-61B – are in fact the two components of the well-known, closely separated visual binary UV Ceti, observed within a single ALMA pointing; the reported flux (0.85 mJy) is for the spatially unresolved, blended pair and cannot be attributed to either component individually. The remaining three detections (τ Ceti, and β Pictoris in both Bands 3 and 6) are consistent with previously known circumstellar/debris material or stellar activity in these well-studied systems and are not treated as anomalous. The population-level anomaly check described in §4.3 has been applied to every target processed to date but has not yet flagged any star, and in most T_{eff} bins the survey does not yet have the minimum of five comparison stars needed for the population-relative component of the test; both the physical-baseline and population-relative statistics will accumulate statistical power as the survey proceeds.

6. DISCUSSION

The results presented here are necessarily preliminary: only 21 of 120 planned target/bands ($\sim 18\%$) have been completed, and the survey is ordered to complete the nearest, most sensitively-constrained systems first, so the even-

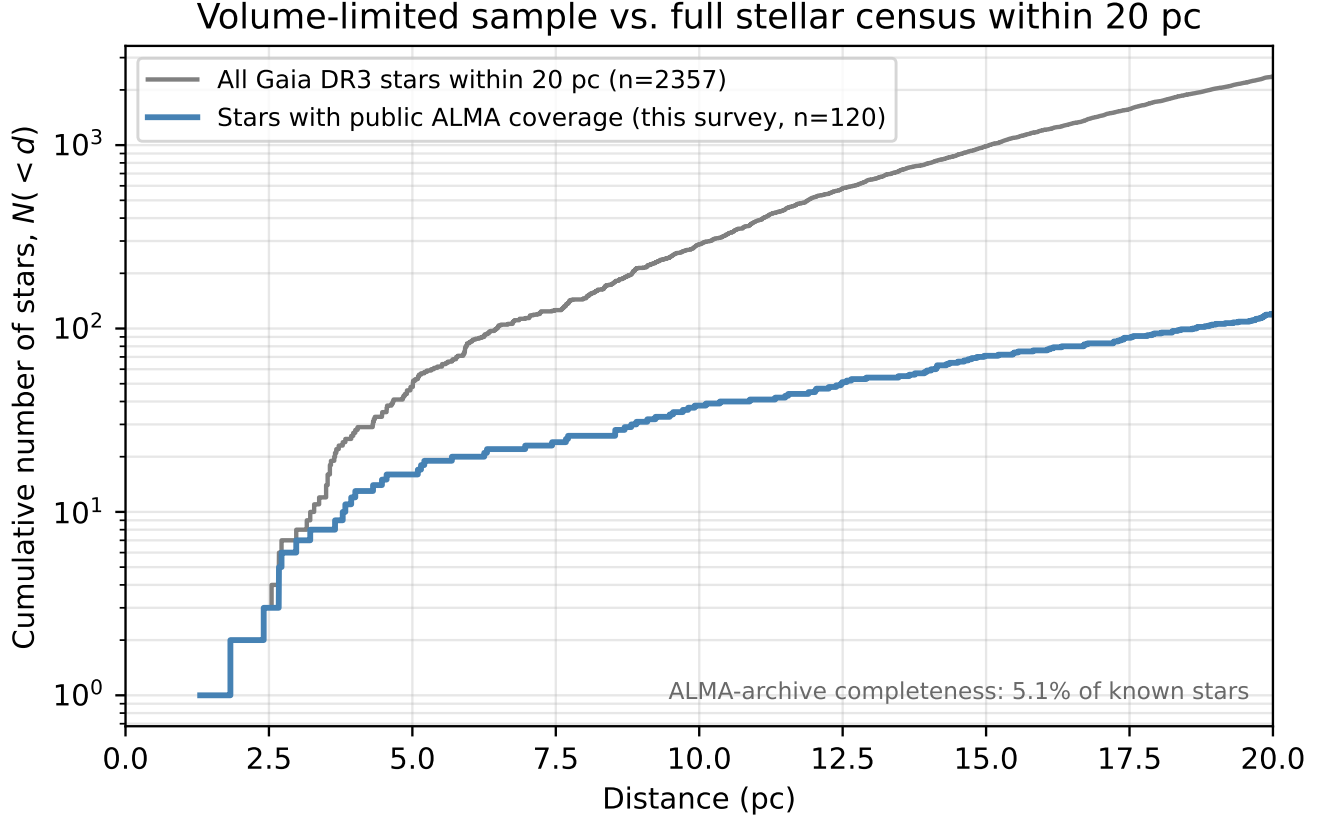


FIG. 2.— Cumulative number of stars, $N(< d)$, within 20 pc as a function of distance d . The grey curve is the full known stellar population within 20 pc from our reference Gaia DR3 catalogue (2357 stars); the blue curve is our 120-star volume-limited sample – every one of those 2357 stars that additionally has public ALMA archival coverage, only $\sim 5\%$ of the total. The two curves are not expected to track one another: ALMA proposal pressure on the immediate solar neighbourhood is itself not spectral-type- or distance-uniform, so our sample is volume-limited and unbiased *within* the ALMA-observed population, but is not a complete or unbiased sample of the local stellar population as a whole (see §3).

tual full sample will extend these results to significantly larger distances and, correspondingly, shallower EIRP and continuum limits at the faint end. No credible technosignature has been found among the systems searched so far; the one automatically-flagged candidate is confidently attributable to CO(1–0) line emission rather than an artificial signal, underlining the importance – already highlighted by Mason et al. (2024) – of checking any millimetre-band narrowband hit against known molecular transitions before treating it as a candidate; this check is now automated for every future hit (§4). Applying the statistical frameworks of Wright et al. (2018) and Margot et al. (2023) to derive a quantitative bound on the prevalence of detectable transmitters in the solar neighbourhood is deferred to a future version of this paper, once a statistically meaningful fraction of the volume-limited sample has been completed; with only 21 systems searched to date such a bound would not yet be usefully constraining.

A pointing-position sanity check introduced during this update (§4) found that four target/bands that otherwise completed processing (SCR J1845–6357, the G 9–38AB pair, and Wolf 358) are associated with an ALMA dataset whose actual pointing lies tens to hundreds of primary-beam-widths from the star’s catalogued position – almost certainly an upstream target-to-archive crossmatch error rather than a real result for that star. One of these (G 9–38AB) had, prior to this check being added, produced a continuum “detection” of order 8 Jy, wildly inconsistent with every other measurement in this survey (all sub-10 mJy) and almost certainly attributable to whatever unrelated bright source that MOUS’s true pointing happens to contain. These four target/bands are excluded from Table 1 and from all statistics quoted here; they are being re-investigated, and the pointing check itself has been added to the pipeline as a standing safeguard against this class of error recurring silently.

The two supplementary analyses introduced in this work (§4.2, §4.3) are best understood as infrastructure investments rather than sources of results in their own right at this early stage. The closure-phase test has not yet been exercised against real data, simply because no candidate other than the already-explained β Pictoris CO line has crossed threshold; it remains available, at negligible ongoing cost, for any future hit. The population-level anomaly check is accumulating the population of same- T_{eff} -bin comparison stars it needs to become statistically powerful, and we expect it to become genuinely diagnostic only once several dozen targets per broad spectral-type bin have been processed. We regard both as a useful methodological contribution independent of whether they ultimately flag anything in this particular sample: to our knowledge, neither has previously been applied in a technosignature search, and both are

made possible specifically by properties of this survey (interferometric data; a volume-limited, unbiased sample) that do not hold for the great majority of prior technosignature work reviewed in §2.

The UV Ceti unresolved-pair case (§5) illustrates a general caveat for any archival, position-agnostic survey of this kind: close visual binaries and multiple systems that are unresolved at the relevant ALMA angular resolution will necessarily share a single measurement, and any future statistical analysis of this sample’s completeness must account for this rather than double-counting such pairs as independent trials.

7. CONCLUSIONS

We have described the design, methodology, and first results of a volume-limited technosignature search of all 120 stars within 20 pc with public ALMA archival coverage, irrespective of disc-hosting or planet-host status. Twenty-one of the planned 120 target/bands have been completed as of this version, yielding EIRP limits of 1.6×10^{13} – 2.8×10^{15} W and sixteen continuum non-detections reaching a median 5σ limit of 0.55 mJy, with no credible technosignature identified. We have introduced two analyses – interferometric closure-phase vetting and a population-level, spectral-type-normalised continuum anomaly check – that exploit properties unique to this survey and are, to our knowledge, new to the technosignature literature; both are now running as a standard part of the pipeline and will accumulate statistical power as the survey proceeds. This paper will be updated as further targets are completed, and we anticipate extending the volume limit to 30, 40, and 50 pc as a natural next step once the 20 pc sample is complete.

ACKNOWLEDGEMENTS

This paper makes use of the following ALMA data: [the full list of ALMA project codes used will be compiled once the survey is complete; see Table 1 in Appendix A for the individual execution blocks searched to date]. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC).

DATA AVAILABILITY

The reduced data products underlying this paper, together with the full target-selection and analysis pipeline, are being made available at <https://github.com/Tilanthi/SETI> as the survey progresses. Raw ALMA visibility data are publicly available from the ALMA Science Archive at <https://almascience.org>.

APPENDIX

FULL PER-TARGET RESULTS TABLE

Table 1 lists the full per-target/band results underlying the summary given in Section 5: spectral type, the narrow-band technosignature frequency range actually searched, the number of spectral windows configured simultaneously by the correlator for that observation (“ N_{spw} ”), the resulting EIRP_{min} limit, and the continuum flux density (or, for non-detections, its 5σ upper limit) together with the underlying map rms noise. EIRP_{min} is the minimum EIRP that would have produced a 5σ detection in that spectral window – i.e. $\text{EIRP}_{\text{min}} = 4\pi d^2 (5\sigma_{\text{comb}}) \Delta\nu$, using the same 5σ threshold, and the same primary-beam correction, as the continuum column – so the two limits in each row are directly comparable and neither is a bare, unnormalised measured value. As described in §4, the pipeline was extended during the preparation of this version to search every distinct configured spectral window (not only the single finest-resolution one, as in earlier versions of this paper), and each searched window appears as its own row where available (e.g. GJ 581, now with all four of its configured windows searched); most of the 19 target/bands processed before this extension are still being reprocessed under the new, multi-window pipeline, so only their originally-searched window is shown for now, explicitly flagged in the Notes column. Two new target/bands (Kapteyn’s Star/HD 33793 and Van Maanen’s Star/Wolf 28) are included for the first time in this version. Four further target/bands that completed processing during this update (SCR J1845–6357, the G 9–38AB pair, and Wolf 358) are deliberately *excluded* from the table: an automated sanity check (§4) found their assigned ALMA dataset’s actual pointing lies tens to hundreds of primary-beam-widths from the star’s catalogued position, indicating an upstream target-MOUS crossmatch error rather than a genuine result for that star; these are being re-investigated and will be reprocessed against a corrected dataset in a future version. This table will continue to grow, and existing rows will be superseded by per-window results, as the survey proceeds; the version current at the time of writing is reproduced here so that the summary statistics quoted in the main text are fully traceable.

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|---|---|

TABLE 1

PER-TARGET/BAND RESULTS, ORDERED BY DISTANCE. BAND NUMBERS ARE GIVEN WITHOUT THE CUSTOMARY “B” PREFIX TO AVOID CONFUSION WITH B-TYPE SPECTRAL CLASSIFICATIONS ELSEWHERE IN THE TABLE. FREQUENCY RANGE IS IN GHz; N_{spw} IS THE TOTAL NUMBER OF SPECTRAL WINDOWS CONFIGURED SIMULTANEOUSLY IN THE OBSERVATION. EIRP_{min} (W) IS THE MINIMUM EIRP THAT WOULD REGISTER AS A 5σ DETECTION IN THAT SPECTRAL WINDOW (§4), PRIMARY-BEAM CORRECTED. FLUX IS THE MEASURED PEAK CONTINUUM FLUX DENSITY FOR A DETECTION, OR THE RESULTING 5σ UPPER LIMIT (DENOTED “ $<$ ”), ALSO PRIMARY-BEAM CORRECTED; RMS IS THE UNDERLYING (POST-CORRECTION) CONTINUUM IMAGE RMS NOISE; BOTH IN mJy. FOR GJ 581, ALL FOUR CONFIGURED SPECTRAL WINDOWS HAVE NOW BEEN SEARCHED (ONE ROW EACH); ITS SINGLE CONTINUUM MEASUREMENT, COMMON TO ALL FOUR WINDOWS, IS GIVEN ONCE, IN THE FIRST ROW. THE ONE AUTOMATICALLY-FLAGGED TECHNOSIGNATURE CANDIDATE IS NOTED DIRECTLY IN THE NOTES COLUMN; SEE §5 FOR ITS DISPOSITION.

Target	Band	Sp. type	Dist. (pc)	Freq. range (GHz)	N_{spw}	EIRP_{min} (W)	Flux (mJy)	RMS (mJy)	Notes
Proxima Cen	6	M6V	1.30	223.09–224.91	4	3.43e+13	<0.666	0.1332	reprocessing (other spws) queued
Barnard’s Star	7	M5V	1.83	...	6	...	<3.622	0.7243	narrowband search failed; reprocessing queued
Wolf 359	7	M6V	2.41	...	7	...	<3.380	0.6761	narrowband search failed; reprocessing queued
Sirius B	3	DA2	2.67	92.13–93.86	4	2.65e+13	<0.196	0.0393	primary-beam corrected ($\times 1.036$, §4); reprocessing (other spws) queued
Sirius B	4	DA2	2.67	142.13–143.86	4	2.53e+13	<0.271	0.0540	primary-beam corrected ($\times 1.096$); reprocessing (other spws) queued
Sirius B	5	DA2	2.67	195.13–196.86	4	4.04e+13	<0.528	0.1057	primary-beam corrected ($\times 1.160$); reprocessing (other spws) queued
G 272-61B	3	M6Ve	2.67	89.63–91.36	4	1.65e+13	0.850	0.0260	unresolved UV Cet AB pair, blended flux; reprocessing (other spws) queued
G 272-61A	3	M5.5Ve	2.72	89.63–91.36	4	1.71e+13	0.850	0.0260	unresolved UV Cet AB pair, blended flux; reprocessing (other spws) queued
CD-23 14742	6	M5V	2.98	223.09–224.91	4	1.20e+14	<0.388	0.0776	reprocessing (other spws) queued
tau Cet	6	K0V	3.65	229.55–231.55	24	5.46e+13	0.914	0.1323	reprocessing (other spws) queued
HD 33793	6	sdM1	3.93	223.09–224.91	4	5.62e+14	<0.578	0.1157	Kapteyn’s Star; reprocessing (other spws) queued
Wolf 28	6	DZ7	4.31	223.13–224.87	4	6.67e+13	<0.161	0.0321	Van Maanen’s Star (white dwarf); reprocessing (other spws) queued
GJ 674	6	M3V	4.55	223.09–224.91	4	2.74e+14	<0.516	0.1032	reprocessing (other spws) queued
GJ 581	6	M4V	6.30	223.01–224.99	4	6.35e+14	<0.451	0.0903	4/4 spws searched, no candidate
GJ 581	6	M4V	6.30	225.01–226.99	4	6.01e+14	—	—	
GJ 581	6	M4V	6.30	239.01–240.99	4	6.61e+14	—	—	
GJ 581	6	M4V	6.30	241.01–242.99	4	6.97e+14	—	—	
GJ 849	6	M4V	8.81	229.08–230.92	4	8.30e+14	<0.712	0.1424	reprocessing (other spws) queued
HD 285968	6	M1V	9.49	229.04–230.88	4	3.45e+14	<0.700	0.1400	reprocessing (other spws) queued
HD 23484	6	K5V	16.17	229.61–231.45	4	1.70e+15	<0.497	0.0995	reprocessing (other spws) queued
HD 10647	7	G0V	17.35	330.35–332.19	4	1.97e+15	<2.020	0.4040	reprocessing (other spws) queued
HD 53143	6	K0V	18.34	229.60–231.44	4	2.84e+15	<0.877	0.1754	reprocessing (other spws) queued
bet Pic	3	F0V	19.63	114.83–115.69	4	2.08e+14	0.444	0.0281	CO(1-0) line, candidate; see text
bet Pic	6	F0V	19.63	230.08–231.00	4	2.01e+15	5.659	0.9626	reprocessing (other spws) queued

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